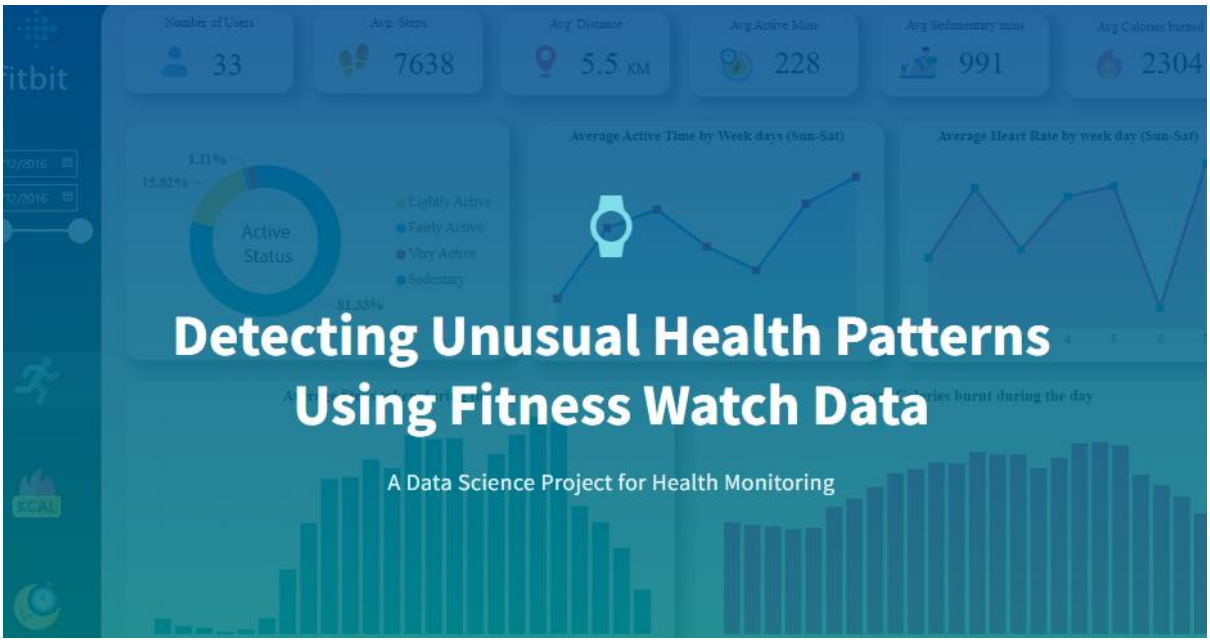
1. **Title: Detecting Unusual Health Patterns Using Fitness Watch Data**



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2. Project Statement:

With the surge in wearable fitness devices, a vast amount of health-related time-series data is generated daily. Yet, users and healthcare providers often fail to notice subtle early warning signs of health anomalies like irregular heartbeats, sleep disturbances, or abnormal activity levels. This project leverages AI-based anomaly detection to intelligently analyze fitness watch data and flag unusual health patterns. By enabling proactive health monitoring, it supports preventive healthcare and personalized wellness insights.

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3. Outcomes:

- Accurate detection of anomalies in heart rate, sleep duration, and step count.
- Personalized health alerts using time-series models and clustering techniques.
- Integration with raw fitness watch data (CSV/JSON format).
- Real-time or batch-based anomaly flagging for dynamic use cases.
- Configurable metric thresholds (e.g., HR > 120 bpm for > 10 mins).
- Interactive dashboards for visual trend and anomaly tracking.
- Exportable reports for users or healthcare professionals.

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4. Modules to be Implemented:

Module 1: Data Collection and Preprocessing

- Import health data (heart rate, steps, sleep) from fitness trackers in CSV/JSON format.
- Clean and normalize timestamps, interpolate missing values.
- Resample to consistent time intervals (e.g., per minute, hourly).

Module 2: Feature Extraction and Modeling

- Extract relevant time-series features using TSFresh (mean, std, kurtosis, etc.).
- Apply Facebook Prophet to model seasonal trends and detect deviations.
- Use clustering (e.g., KMeans, DBSCAN) to group similar behavioral patterns.

Module 3: Anomaly Detection and Visualization

- Rule-based (thresholds) and model-based anomaly detection (Prophet residuals, clustering outliers).
- Visualization with Matplotlib or Plotly: Line charts, anomaly highlights, time windows.

Module 4: Dashboard for Insights

- Build an interactive dashboard using Streamlit.
- Allow users to upload fitness data files, trigger anomaly detection dynamically.
- Generate downloadable reports for anomaly summaries and trends.

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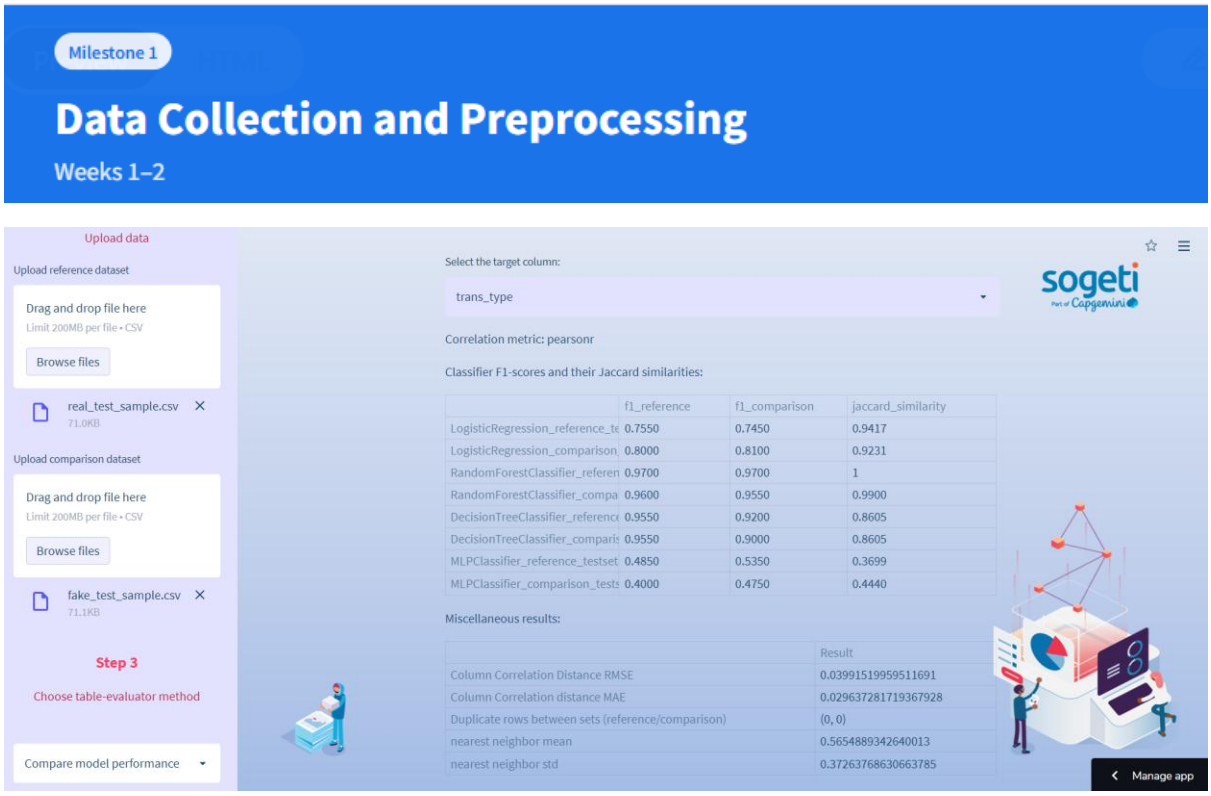
5. Week-wise Module Implementation and High-Level Requirements:

Milestone 1 (Weeks 1–2): Data Collection and Preprocessing

- Fitness data (CSV/JSON) ingestion logic.
- Normalize timestamps (UTC), handle missing/null values.
- Align time intervals to consistent frequency (e.g., 1-min granularity).

Output Screenshots:

- File upload UI
- Cleaned dataset preview
- Time-normalized data log



Milestone 2 (Weeks 3–4): Feature Extraction and Modeling

- Extract statistical features using TSFresh.
- Trend modeling with Facebook Prophet on HR/sleep/steps.
- Apply clustering (KMeans, DBSCAN) to detect behavior patterns.

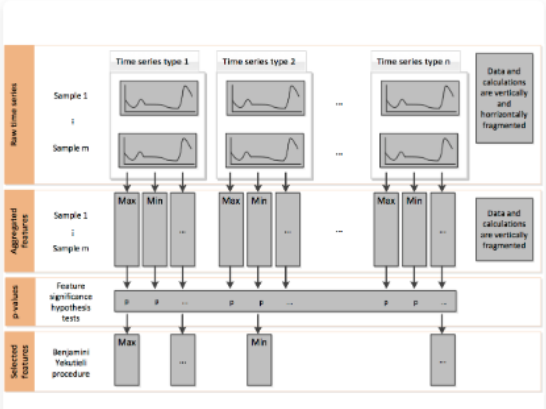
Output Screenshots:

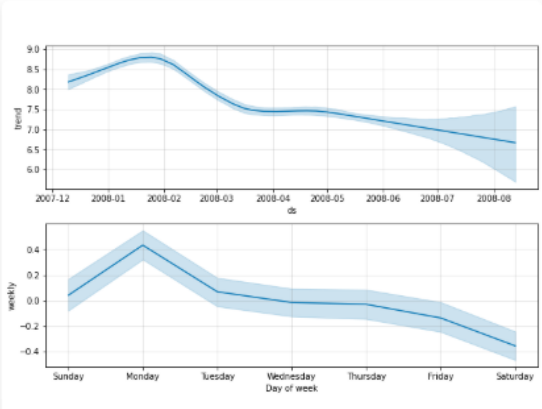
- TSFresh feature matrix
- Prophet trend graph (with confidence intervals)
- Clustering result plot (PCA/t-SNE projection)

Milestone 2

Feature Extraction and Modeling

Weeks 3–4





Milestone 3 (Weeks 5–6): Anomaly Detection and Visualization

- Flag anomalies via:
  - Model residuals
  - Threshold violations
  - Outlier clusters
- Show anomalies on interactive charts (with annotations).

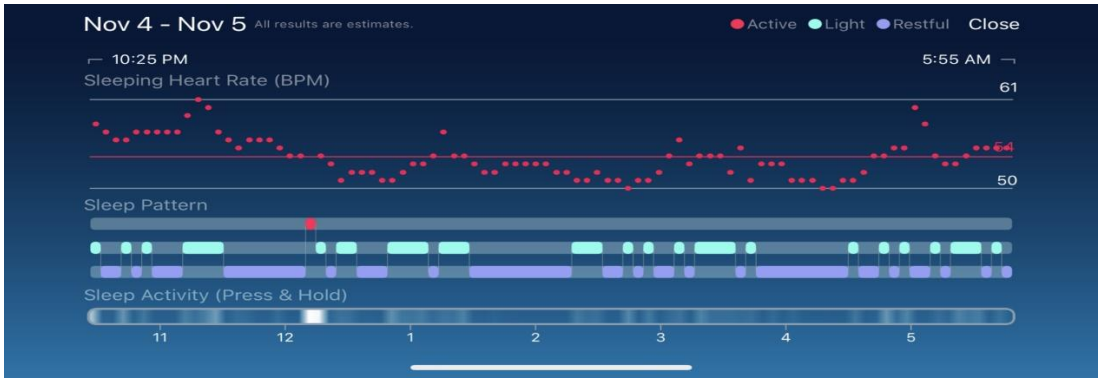
Output Screenshots:

- Heart rate chart with anomalies highlighted
- Sleep pattern visualization
- Step count trend with alerts

Milestone 3

Anomaly Detection and Visualization

Weeks 5–6

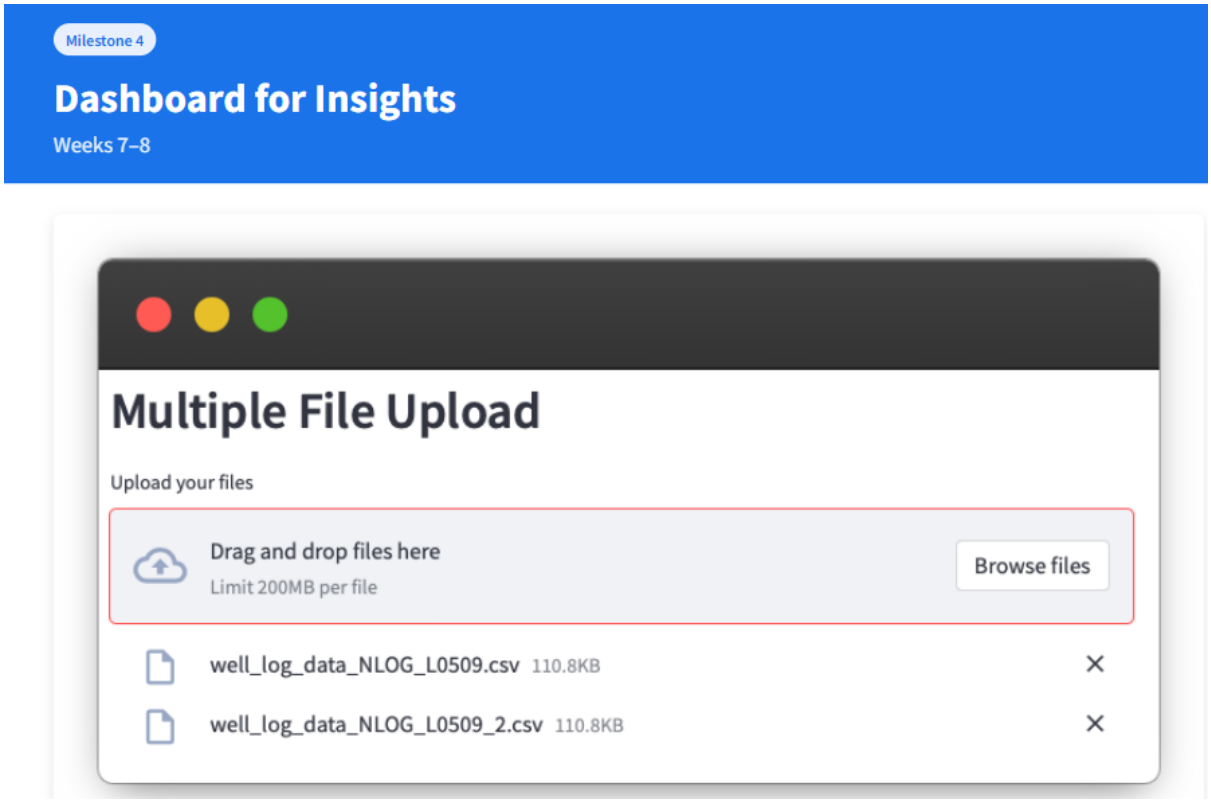


Milestone 4 (Weeks 7–8): Dashboard for Insights

- Streamlit-based interactive UI:
  - File upload
  - Trigger anomaly detection
  - Filter by date/metric
- Export insights in PDF/CSV format

Output Screenshots:

- Streamlit dashboard UI
- Downloadable report button



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6. Evaluation Criteria:

Milestone 1 Evaluation (Week 2):

- Successfully imports and cleans data (CSV/JSON).
- Consistent time-series format ready for modeling.

Milestone 2 Evaluation (Week 4):

- TSFresh features correctly extracted.
- Prophet trends visible for all metrics.
- Clustering models running without errors.

Milestone 3 Evaluation (Week 6):

- 90%+ simulated anomalies are accurately detected.
- Plots and anomaly flags visually align with data patterns.

Milestone 4 Evaluation (Week 8):

- Streamlit dashboard functions correctly.
- Users can upload files, view alerts, and download reports.

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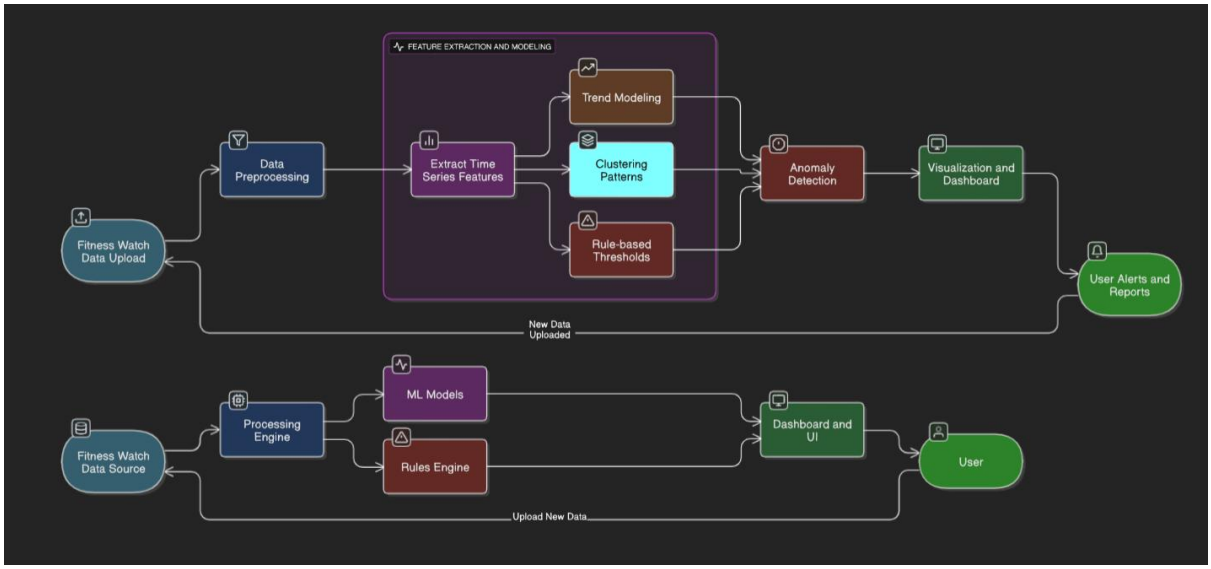
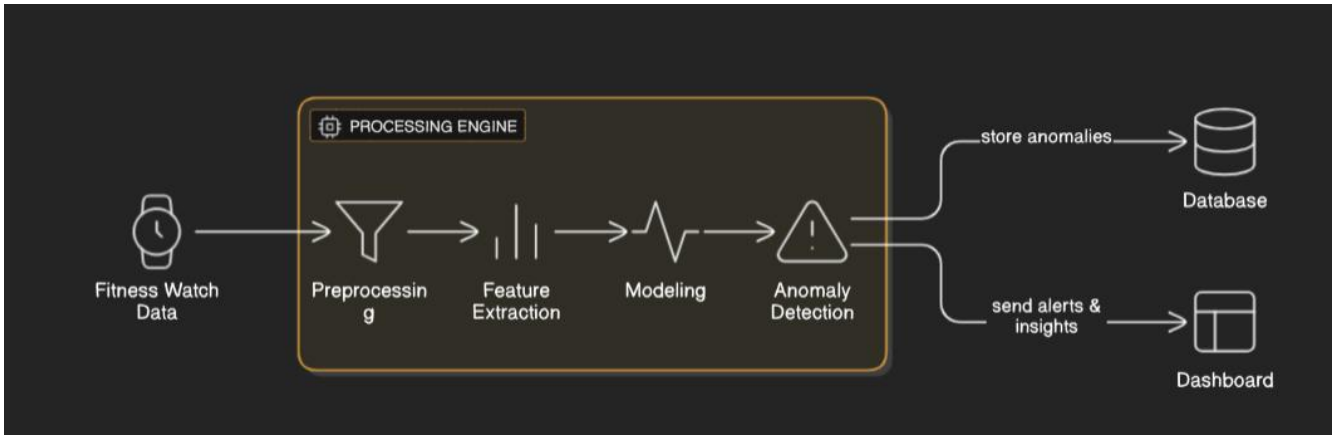
7. Workflow Diagram:

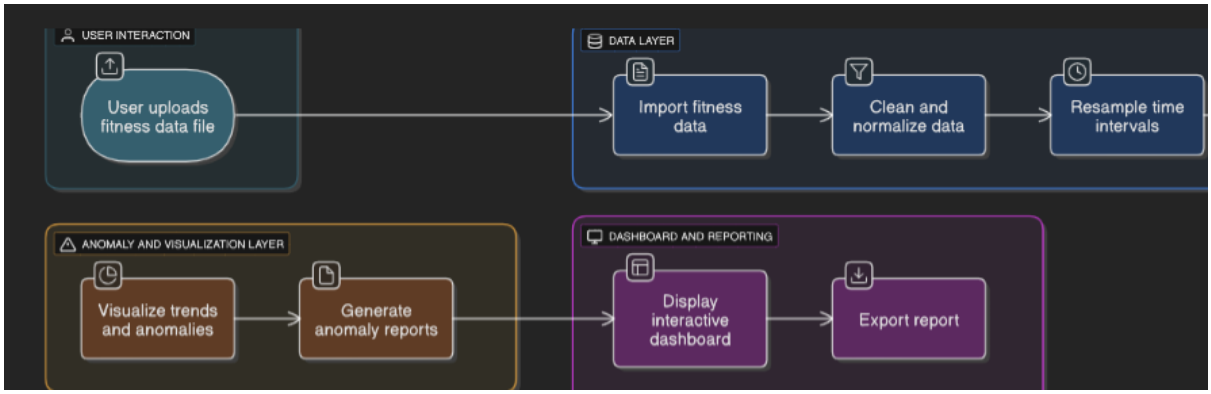
1. Data Import → 2. Preprocessing → 3. Feature Extraction →
2. Modeling → 5. Anomaly Detection → 6. Visualization & Dashboard

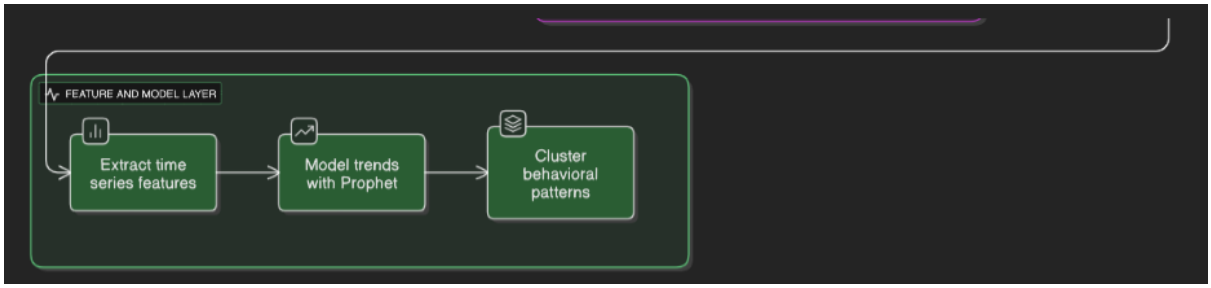


8. Architecture Diagram:



9.Class Diagram




FEATURE AND MODEL LAYER
Extract time series features
Model trends with Prophet
Cluster behavioral patterns